MODULE 02 · DOMAIN 1 - SNOWFLAKE AI DATA CLOUD FEATURES AND ARCHITECTURE

Editions, and the 'minimum edition' question

Less a topic than a question format. Four tiers, and the boundaries the exam actually asks about.

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Editions are **cumulative**. So the answer is never 'which edition has it', it is **'where does it first appear'**.

The four tiers, in one line each

EDITION	WHAT IT IS FOR
Standard	The introductory tier. Full, unlimited access to all standard features.
Enterprise	Large-scale organisations. Adds performance features and the governance policy family.
Business Critical	Extremely sensitive data. Adds encryption control, private connectivity, compliance, and failover.
Virtual Private Snowflake	The strictest requirements. A completely separate environment sharing no hardware with other accounts.

ON THE EXAM

Business Critical was formerly called Enterprise for Sensitive Data. Older material still uses the old name.

What Standard already includes

The ones people get wrong, by assuming security must mean Enterprise

- **Federated authentication and SSO. OAuth. MFA. Network policies.** All four are Standard.
- **Fail-safe, 7 days.** Every edition. Only Time Travel is tiered.
- **Replication** of databases and shares between accounts in an organisation.
- **Dynamic tables, external tables, hybrid tables, Iceberg tables.** Clustering keys and Automatic Clustering.
- **Snowpipe, Snowpipe Streaming, Kafka connector, streams, tasks.**

> ON THE EXAM
>
> If a question implies you must upgrade to get authentication, encryption or continuous ingestion, distrust it.

Standard also includes all of this

- **The entire Cortex surface.** AI functions, Copilot, Cortex Analyst, Cortex Search, Cortex Fine-tuning.
- **All of Snowflake ML.** ML Functions, Model Registry, Feature Store.
- **Snowpark and Streamlit in Snowflake.** External functions and external network access.
- **Marketplace, public and private listings**, and cross-cloud auto-fulfilment.
- **Snowflake Optima**, at no extra cost, though it needs a Gen2 or Adaptive warehouse.

> ON THE EXAM
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> None of the AI content is edition-gated. A scenario cannot force an upgrade by mentioning Cortex.

What Enterprise adds

CLUSTER	FEATURES
Time Travel	Extended retention, up to 90 days. Standard is capped at 1 day.
Performance	Multi-cluster warehouses. Materialized views. Search optimization. Query acceleration.
Governance	Masking. Row access. Aggregation and projection policies. Differential privacy.
Audit	Classification. ACCESS_HISTORY view. Data metric functions.
Other	Periodic rekeying. Synthetic data. Your own clean rooms. Early release access.

ON THE EXAM

Two clusters do most of the work: the four accelerators, and the policy family.

What Business Critical adds

CLUSTER	FEATURES
Encryption	Tri-Secret Secure. Customer-managed keys plus Snowflake-managed keys.
Networking	PrivateLink, Azure Private Link, Google Private Service Connect.
Continuity	Failover and failback between accounts. Client redirect.
Compliance	PHI under HIPAA and HITRUST, needing a signed BAA. PCI DSS. FedRAMP. ITAR.

ON THE EXAM

Four signals: customer-managed keys, private networking, failover, a named regulation.

Virtual Private Snowflake, and the quirk

- **A dedicated metadata store and a dedicated pool of compute.** No hardware shared with accounts outside the VPS.
- Full encryption of **all** data transmissions, including internal ones inside the isolated environment.
- The **account locator format differs** from every other edition.
- **Marketplace and public listings are not available.** Private listings, Universal Search and cross-cloud auto-fulfilment are.
- VPS accounts cannot collaborate in Snowflake Data Clean Rooms, though they can create their own.

> ON THE EXAM
>
> VPS is the one tier that is not strictly a superset. Isolation costs reach, which is the whole point of it.

The trap the exam sets most often

STANDARD IS ENOUGH

Replication

- Keep a synchronised copy in another region
- Databases and shares, within one organisation
- Read-only secondary you can promote manually
- Available in every edition

BUSINESS CRITICAL REQUIRED

Failover and failback

- Automatic promotion of the secondary
- Client redirect, so applications follow
- Failover groups for coordinated recovery
- Business Critical or higher, no exceptions

THE TIEBREAKER

Read the verbs. 'Keep a copy' is replication. 'Fail over automatically' or 'redirect clients' is Business Critical.

Minimum edition: the greatest hits

IF THE SCENARIO NEEDS...	MINIMUM EDITION
Time Travel beyond 1 day, up to 90	Enterprise
Multi-cluster warehouse, materialized view, search optimization, query acceleration	Enterprise
Masking policy, row access policy, classification, ACCESS_HISTORY	Enterprise
Customer-managed keys, private connectivity, automatic failover	Business Critical
PHI, PCI, FedRAMP, or any named regulation	Business Critical
No shared hardware, dedicated metadata store	Virtual Private Snowflake

ON THE EXAM

Six rows will answer almost every minimum-edition question on the paper. Learn these before anything else in this module.

How to answer one under time pressure

- **Find the named feature first.** The scenario dressing rarely matters. One noun decides it.
- **If two features are named, the higher one wins.** Masking plus PHI is Business Critical, not Enterprise.
- **Do not upgrade for security basics.** SSO, OAuth, MFA and network policies are Standard.
- **Do not upgrade for AI.** Cortex and Snowflake ML are Standard throughout.
- If nothing in the list appears, the answer is probably **Standard**, and that is a real answer.

> ON THE EXAM
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> The commonest error is answering one tier too high. Snowflake is more generous at Standard than most candidates expect.

> A hospital group is designing a Snowflake account. Patient records containing protected health information will be stored and analysed. Analysts must see a masked version of the patient identifier, while a small clinical role sees the full value.
>
> The compliance team also requires that the organisation controls one of the encryption keys protecting the data at rest.

What is the minimum Snowflake edition required?

A Standard, because column-level security and encryption are included in all editions.

B Enterprise, because masking policies are an Enterprise feature.

C Business Critical, because of the PHI and customer-managed keys.

D Virtual Private Snowflake, because health data requires dedicated infrastructure.

◆ PAUSE AND DECIDE

What is the minimum Snowflake edition required?

A. Standard.

Encryption is universal, but column-level security is not. Masking policies start at Enterprise.

B. Enterprise.

Correct about masking, and it stops reading. PHI and customer-managed keys both push higher.

C. Business Critical.

PHI needs HIPAA and HITRUST support, and Tri-Secret Secure delivers the customer-managed key. Both are Business Critical.

D. Virtual Private Snowflake.

Would work, and nothing asked for isolated hardware. Minimum means minimum, so this overshoots.

THE SIGNAL

Two requirements, both Business Critical. The masking detail was there to make Enterprise look sufficient.

> A retailer runs its production account on Standard Edition in one region. After an outage at its cloud provider, the board asks for a disaster recovery plan.
>
> The requirement is that a copy of the production databases exists in a second region at all times, and that if the primary region is lost, the secondary is promoted and client applications reconnect to it without being reconfigured.

What must change, if anything?

A. Upgrade to Business Critical Edition, which adds failover, failback and client redirect.

B. Upgrade to Enterprise Edition, which adds failover and client redirect.

C. Nothing. Replication, failover and client redirect are all available in Standard Edition.

D. Nothing. Use zero-copy cloning into a second region on a scheduled task instead.

◆ PAUSE AND DECIDE

What must change, if anything?

A Upgrade to Business Critical.

Failover, failback and redirecting client connections all first appear at Business Critical.

B Upgrade to Enterprise.

Enterprise adds performance and governance features. It adds nothing to business continuity.

C Nothing. All three are in Standard.

Half right. Replication is Standard, but failover, failback and client redirect are not.

D Scheduled zero-copy clones into a second region.

A clone cannot cross regions. Cloning is metadata pointing at micro-partitions that live in one region's storage.

THE SIGNAL

The first half of the requirement is Standard. The second half is not. Requirements often split across a boundary like this.

A data team on Standard Edition wants to build a natural-language interface over its sales tables using Cortex Analyst, deploy a Streamlit app for the results, and refresh the underlying tables automatically with dynamic tables.

Their account manager has suggested that this work requires an Enterprise upgrade. Finance has asked for that claim to be checked.

Is an upgrade required, and why?

A. Yes. Cortex Analyst and Cortex Search both require Enterprise Edition.

B. Yes. Dynamic tables require Enterprise, because they are a form of materialized view.

C. Yes. Streamlit in Snowflake requires Enterprise because it runs application code.

D. No. Cortex Analyst, Streamlit in Snowflake and dynamic tables are all available in Standard.

◆ PAUSE AND DECIDE

Is an upgrade required, and why?

A Yes, Cortex requires Enterprise.

The entire Cortex surface is Standard. No AI feature is edition-gated.

B Yes, dynamic tables are a materialized view.

Materialized views are Enterprise, and dynamic tables are a different object and are Standard.

C Yes, Streamlit runs application code.

Streamlit in Snowflake, Snowpark and external access are all Standard.

D No. All three are available in Standard.

Correct on all three, and 'no upgrade needed' is a real answer the exam does use.

THE SIGNAL

B is the best distractor here. Dynamic tables and materialized views both maintain results automatically, and only one is Enterprise.

> A financial institution runs a Virtual Private Snowflake account for its most sensitive analytics. A business unit wants to publish a curated dataset as a public listing on the Snowflake Marketplace to generate revenue.
>
> An architect has assumed this is possible, on the grounds that VPS includes everything in Business Critical.

What should the architect be told?

A. It works as assumed. VPS is a strict superset of Business Critical.

B. Marketplace and public listings are not available in VPS. Private listings are.

C. It works, but only after enabling cross-cloud auto-fulfilment, which VPS disables by default.

D. It works only if the listing is published from a reader account created by the VPS account.

◆ PAUSE AND DECIDE

What should the architect be told?

A VPS is a strict superset.

The stated assumption, and the official matrix contradicts it. VPS trades reach for isolation.

B No Marketplace or public listings in VPS. Private listings are available.

Matches the matrix. Public reach conflicts with the isolation VPS exists to provide.

C Enable cross-cloud auto-fulfilment first.

Auto-fulfilment is available in VPS, and it is not the blocker. Invented configuration.

D Publish from a reader account.

A reader account is a consumer-side mechanism for people without a Snowflake account. It does not publish listings.

THE SIGNAL

'Higher edition' is not the same as 'more capable'. VPS buys isolation, and isolation costs reach.

What to carry into the next module

- **Cumulative tiers.** The question is always where a feature *first* appears, never which tier has it.
- **Enterprise:** 90-day Time Travel, the four accelerators, the whole policy family, classification and ACCESS_HISTORY.
- **Business Critical:** customer-managed keys, private connectivity, failover and failback, named regulations.
- **Standard is generous.** SSO, MFA, network policies, replication, Fail-safe, all table types, all of Cortex.
- **VPS is not a strict superset.** No Marketplace, no public listings.

> ON THE EXAM
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> Next: micro-partitions and clustering. The storage internals that most performance questions actually resolve to.